Data paper

Multimodal annotated daily life Cantonese speech dataset for more advanced AI training

Author: Tse Won Huen[a] and Cheung Yu Lung[b]

Faculty of Humanities, The Education University of Hong Kong, Hong Kong SAR, China

Corresponding author: Won Huen, WH, Tse; s1167175@s.eduhk.hk
Yu Lung, YL, Cheung; s1160587@s.eduhk.hk

Author roles

Conceptualization	Identifying issues, problems. Developing research framework, setting research goals and aims	Tse Won Huen Cheung Yu Lung
Data Curation	Processing, cleaning, formatting and rectifying transcripts of the 3 videos	Tse Won Huen
	Cataloguing, readme file	Tse Won Huen Cheung Yu Lung
	Annotating	Cheung Yu Lung
Formal Analysis		Tse Won Huen Cheung Yu Lung
Resources	Identifying suitable study materials: 3 YouTube videos	Tse Won Huen
Writing	Writing initial draft of data paper	Tse Won Huen Cheung Yu Lung

Abstract:

The data covers Cantonese usage in daily life in Hong Kong, including topics of going to markets, seafood, dining, etc. The database sources include 3 sets of transcripts which are processed from 3 YouTube videos and then annotated, focusing on emotional words specific for Cantonese, in particular discourse particles.

The dataset DOI will be assigned upon upload to GitHub/HuggingFace to serve as a free reference and groundwork for future research on Cantonese. During the data collection process, we utilized Clipto to generate transcript files for the YouTube videos and then edited them to become 100% Cantonese transcripts.

Part 1: README.md

Place this in the root of our GitHub/Hugging Face repository.

Markdown

```
# CAN-DAILY: A Multimodal Annotated Dataset of Spontaneous Hong
Kong Cantonese Daily Life Speech

**Project Title:** Creating a Multimodal Annotated Daily Life
Cantonese Speech Dataset for Advanced AI Training
**Group Members:** Cheung Yu Lung (11605879), Tse Won Huen
(11671751)
**Course:** LIN3046 Computational Linguistics (2025-2026)

## Abstract
CAN-DAILY addresses the scarcity of spontaneous, real-life
Cantonese speech resources. While existing corpora focus on
formal broadcasts, this dataset captures natural interactions in
Hong Kong street markets and vlogs. It contains over 1 hour of
speech with utterance-level timestamps, speaker identification,
and three layers of multimodal annotation: emotion (e.g.,
Excited, Happy), dialogue acts (e.g., Claim, Asking), and
conversational targets. The dataset fills a critical gap for
developing "human-like" AI, supporting improvements in Automatic
Speech Recognition (ASR), sentiment analysis, and the study of
Hong Kong-style code-switching.
```

Dataset Specifications

- **Language:** Spontaneous Hong Kong Cantonese (with English code-switching).
- **Domain:** Street market negotiations, family vlogs, and food commentary.
- **Duration:** 1 hour+ (500+ utterances).
- **Format:** Raw WAV (16kHz), processed CSV, and JSON metadata.
- **Annotation Layers:** - Orthographic Transcription (Colloquial Cantonese characters).
  - Emotion Labels (6 categories).
  - Dialogue Acts (Adapted from SWBD-DAMSL).
  - Interlocutor Targets (e.g., Daughter to Dad/Audience).

Repository Structure

```
├── data/
│ ├── raw/ # Original audio/video source references
│ └── processed/ # Cleaned transcripts (CSV/JSON)
├── docs/
│ ├── annotation_guidelines.md
│ ├── metadata_schema.json
│ └── data_paper.pdf # The final manuscript
├── scripts/ # Data cleaning and analysis scripts
└── README.md
```

Access & License

This dataset is released under the **CC BY-NC 4.0** license for research and educational purposes.

Title: CAN-DAILY: A Multimodal Annotated Dataset of Spontaneous Hong Kong Cantonese Daily Life Speech

Authors: Cheung Yu Lung, Tse Won Huen

Affiliation: The Education University of Hong Kong, Hong Kong

1. Overview

Over 80 million people worldwide speak Cantonese. It is an official language in Hong Kong, a dominant language in Macau, and a main dialect in Guangzhou. However, Cantonese remains a low-resourced language in AI and speech technology. Spoken Cantonese is usually transcribed into written Chinese. Existing datasets are often conducted in formal and controlled settings, not in everyday life conversations. There is a shortage of Hong Kong-style Cantonese English code-switching, as well as lack of rich annotations (emotion, conversation analysis, rhythm/prosodies) since only a few regions speak Cantonese, and Cantonese resources for speech and language AI development is lacking. Even though there is Accurate Automatic Speech Recognition (ASR), these AI tools often fail to reflect how Cantonese is actually spoken in Hong Kong today.

Our project aims to create a multimodal annotated daily life Cantonese speech dataset for more advanced AI training, with the target being real-life daily interactions and specialized domains. We would also use other aspects such as emotion labelling, conversation analysis, and rhythm labelling.
Repository location:
Context
The data collected as part of a study on the usage of Cantonese discourse in daily life.

Repository location
Dataset DOI to be assigned upon upload to GitHub/HuggingFace
Context
The data collected as part of a study on the usage of Cantonese discourse particles for expressing emotion in daily life conversations. It supports studies of cultural reflections of Cantonese and research on pragmatic markers.

2. Methods

To produce the dataset, 3 YouTube videos were identified, and 3 sets of transcripts are generated using Clipto. Due to the limitation of the App, substantial editing and rectifying of the transcripts are conducted to make them become 100% Cantonese

transcripts before annotation.

Sampling strategy – We sampled over one hour of unscripted videos from YouTube (vlogs/street interviews) capturing authentic market haggling, seafood in Hong Kong, dining and family dynamics, which are informal settings with representation of middle age (around 30-40), the elderly and of different gender.

Quality control – Transcripts generated by Clipto are checked against the videos and editing was performed.

Processing- Segmentation was performed at natural pause boundaries. Transcriptions use colloquial Cantonese characters, especially discourse particles (e.g., 嘅, 咗) to preserve linguistic authenticity.

Annotation- Annotators perform a multi-pass process. Code-switching is used for annotation such as combining Cantonese with English.

Diarization - Identifying speakers and their intended targets.

Emotional Tagging - Labelling emotions such as excited, happy, confident and apologetic etc. based on prosodies.

Dialogue Acts - Categorizing communicative intent (Claim, Asking, Directive).

3. Dataset Description

Repository name – HuggingFace

Object name – Typically the name of the file or file set in the repository.

Format names and versions – E.g., ASCII, CSV, Autocad, EPS, JPEG, Excel, SQL,

etc.

Creation dates – 2026-04-25

Dataset creators – Tse Won Huen and Cheung Yu Lung (see author roles)

Language – Cantonese is the languages used in the dataset

License – CC

Publication date – 2026-04-25

The dataset is structured as a CSV with the following Columns: LineID, Timestamp,

Transcription, Speaker, Target, Emotion, and Dialogue Act.

Table 1: Sample of Annotated Data

ID	Speaker	Line	Target	Emotion	Act
1.5	Seller	呢啲真係好好食㗎 (These are really delicious)	Dad	Happy	Declaration
1.14	Dad	訂閱！ (Subscribe!)	Audience	Excited	Directive

4. Reuse Potential

On the technical side, the dataset helps enhance the ASR for spontaneous speech and code-switching.

Linguistically, it helps analyse pragmatic particles and emotional prosody in HK Cantonese.

From the educational aspect, it provides authentic materials for Cantonese as a Second Language (CSL) learners.

Educational: Provides authentic materials for Cantonese as a Second Language (CSL) learners.

Feasibility is high by using publicly available/fair-use materials combined with manual & semi-automated annotation. It also ensures high quality while keeping the workload manageable.

However, the sample size is small as only involves transcripts of 3 videos and not all daily life conversations and different age groups are covered.

5. References

Lee, T., Lo, W. K., Ching, P. C., & Meng, H. (2002). Spoken language resources for Cantonese speech processing. Speech Communication, 36(3-4), 327-342.
Li, L., et al. (2026). Wenetspeech-yue: A large-scale Cantonese speech corpus with multi-dimensional annotation. Proceedings of the AAAI Conference on Artificial Intelligence.
Yu, T., et al. (2022). Automatic Speech Recognition Datasets in Cantonese: A Survey and New Dataset. Proceedings of LREC 2022, 6487–6494.

AI Declaration – Clipto is used to generate transcripts of 3 YouTube videos.